

MOBILE ROBOTICS LAB REPORT

Lab Session 8: Building and Visualizing a Custom Mobile Robot using URDF

1. Objective

The objective of this lab session was to construct a mobile robot model using the **Unified Robot Description Format (URDF)**. Focus was placed on designing the base structure of a differential drive robot, understanding link-joint hierarchies, and visualizing the robot within the ROS 2 environment using RViz.

2. Learning Outcomes

- Gained an understanding of the structure and purpose of URDF files for robot description.
- Successfully created and organized a ROS 2 package dedicated to robot description.
- Built a functional mobile robot model utilizing various links and joints.
- Visualized the custom robot model in RViz and inspected coordinate transformations.

3. Methodology & Tools

The following tools and packages were utilized during the lab implementation:

- **URDF**: XML format used to describe the physical configuration (links, joints, sensors).
- **urdf_tutorial**: ROS 2 package providing launch setups for visualization.
- **tf2_tools**: Used for visualizing and inspecting coordinate frame transforms.
- **robot_state_publisher**: Published the state of the robot to the transform tree.

4. Implementation

Task 1: Package Structure

A dedicated folder structure was created within the ROS 2 workspace to house the robot description:

```
~/ros2_ws/src/my_robot_description/  
├─ urdf/  
├─ launch/  
└─ rviz/
```

Task 2: URDF Development

The `my_robot.urdf` file was developed to define the robot's physical links (`base_link`, `camera`) and their relationships (joints).

```
<robot name="my_mobile_robot">  
  <link name="base_link">  
    <visual>  
      <geometry>  
        <box size="0.4 0.3 0.1"/>  
      </geometry>  
      <material name="blue"/>  
    </visual>  
  </link>  
  
  <link name="camera">  
    <visual>  
      <geometry>  
        <cylinder length="0.05" radius="0.05"/>  
      </geometry>  
      <material name="black"/>  
    </visual>  
  </link>  
  
  <joint name="base_camera_joint" type="fixed">  
    <parent link="base_link"/>  
    <child link="camera"/>
```

```
<origin xyz="0.15 0 0.05" rpy="0 0 0"/>
</joint>
</robot>
```

5. Customization & Student Tasks

Custom Features Added:

- **Design:** Implemented a custom chassis with specific dimensions to suit a differential drive configuration.
- **Geometry:** Used a combination of boxes for the body and cylinders for sensor mounts.
- **Joints:** Configured fixed joints for sensors and prepared the model for continuous joints for future wheel integration.
- **Transformations:** Calculated and applied specific XYZ offsets to align the camera sensor at the front edge of the base.

6. Conclusion

The lab successfully demonstrated the workflow for creating a robot description in ROS 2. By defining links and joints in URDF, we established a foundation for both visualization in RViz and future physics-based simulation in Gazebo. The TF tree was verified to ensure that all coordinate frames were correctly linked to the `base_link`.